

RNN Vanishing Gradient

Vanishing Gradients Problem

The brown and black dog, which was playing with the cat, was a german shepherd.

(x₂)

(x₄)

(x₅)

(x₁₄)

(x₁₅)

Function Dependencies with respect to W_X

$$L_t = -y_t \log(\hat{y}_t) \rightarrow \hat{y}_t = \text{softmax}(z_t) \rightarrow z_t = W_Y h_t \rightarrow h_t = \tanh(W_H h_{t-1} + W_X X_t) \quad (1)$$

Chain rule with respect to W_x

$\frac{\partial L_t}{\partial W_x} = \frac{\partial L_t}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial z_t} \frac{\partial z_t}{\partial h_t} \frac{\partial h_t}{\partial W_x}$ but note, that within h_t , h_{t-1} also contains W_x , thus we need to recursively chain rule h_{t-1} until we reach h_0

$$\frac{\partial L_t}{\partial W_x} = \frac{\partial L_t}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial z_t} \frac{\partial z_t}{\partial h_t} \frac{\partial h_t}{\partial W_x} + \frac{\partial L_t}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial z_t} \frac{\partial z_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial W_x} + \dots + \frac{\partial L_t}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial z_t} \frac{\partial z_t}{\partial h_t} \frac{\partial h_t}{\partial h_{t-n}} \frac{\partial h_{t-n}}{\partial W_x} = \sum_{k=0}^t \frac{\partial L_t}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial z_t} \frac{\partial z_t}{\partial h_t} \frac{\partial h_t}{\partial h_k} \frac{\partial h_k}{\partial W_x} \quad (3)$$

Gradient for W_X

$$\frac{\partial L_{total}}{\partial W_X} = \sum_{t=1}^n \sum_{k=0}^t \frac{\partial L_t}{\partial \hat{y}_t} \frac{\partial \hat{y}_t}{\partial z_t} \frac{\partial z_t}{\partial h_t} \frac{\partial h_t}{\partial h_k} \frac{\partial h_k}{\partial W_x} \quad (4)$$

Vanishing Gradients Problem

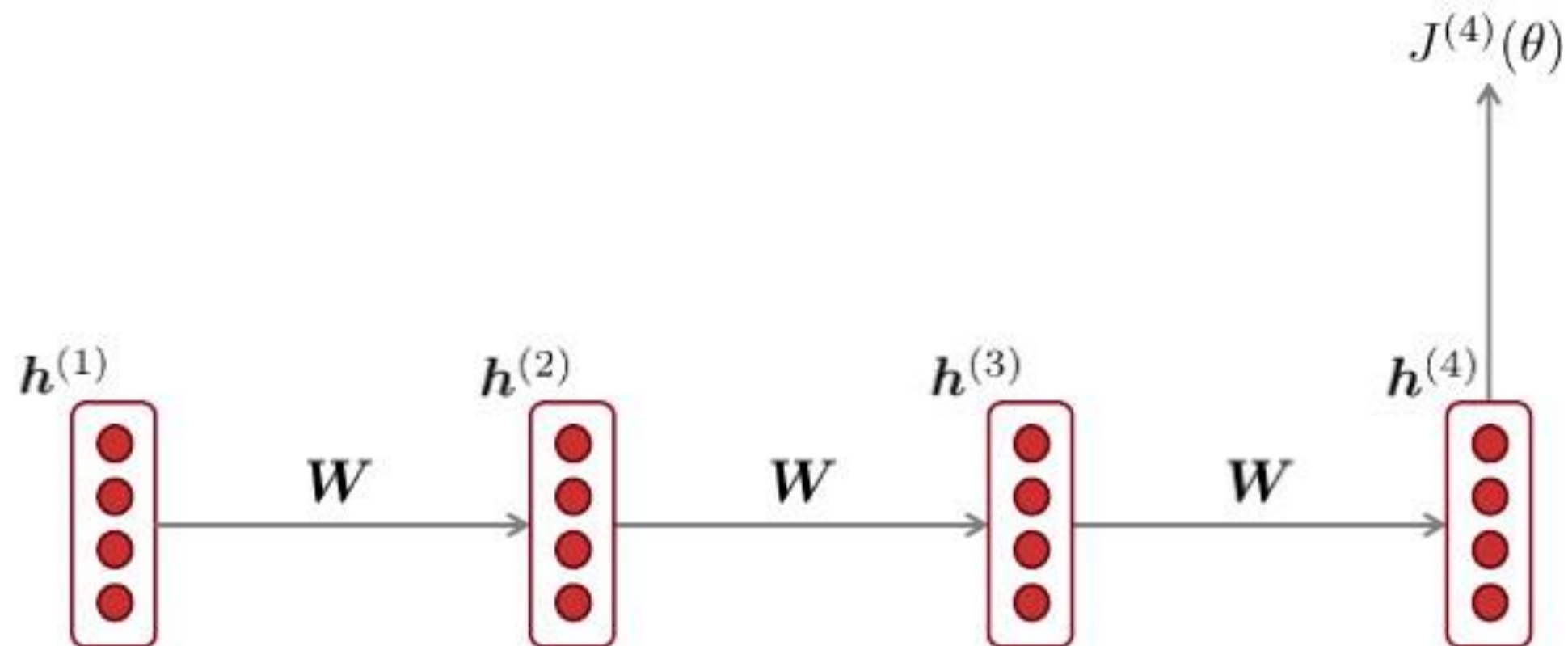
backpropagation error of the word “shepherd” (x15) back to “brown” (x2)

$$\frac{\partial L_{15}}{\partial \hat{y}_{15}} \frac{\partial \hat{y}_{15}}{\partial z_{15}} \frac{\partial z_{15}}{\partial h_{15}} \frac{\partial h_{15}}{\partial h_2} \frac{\partial h_2}{\partial W_x}$$

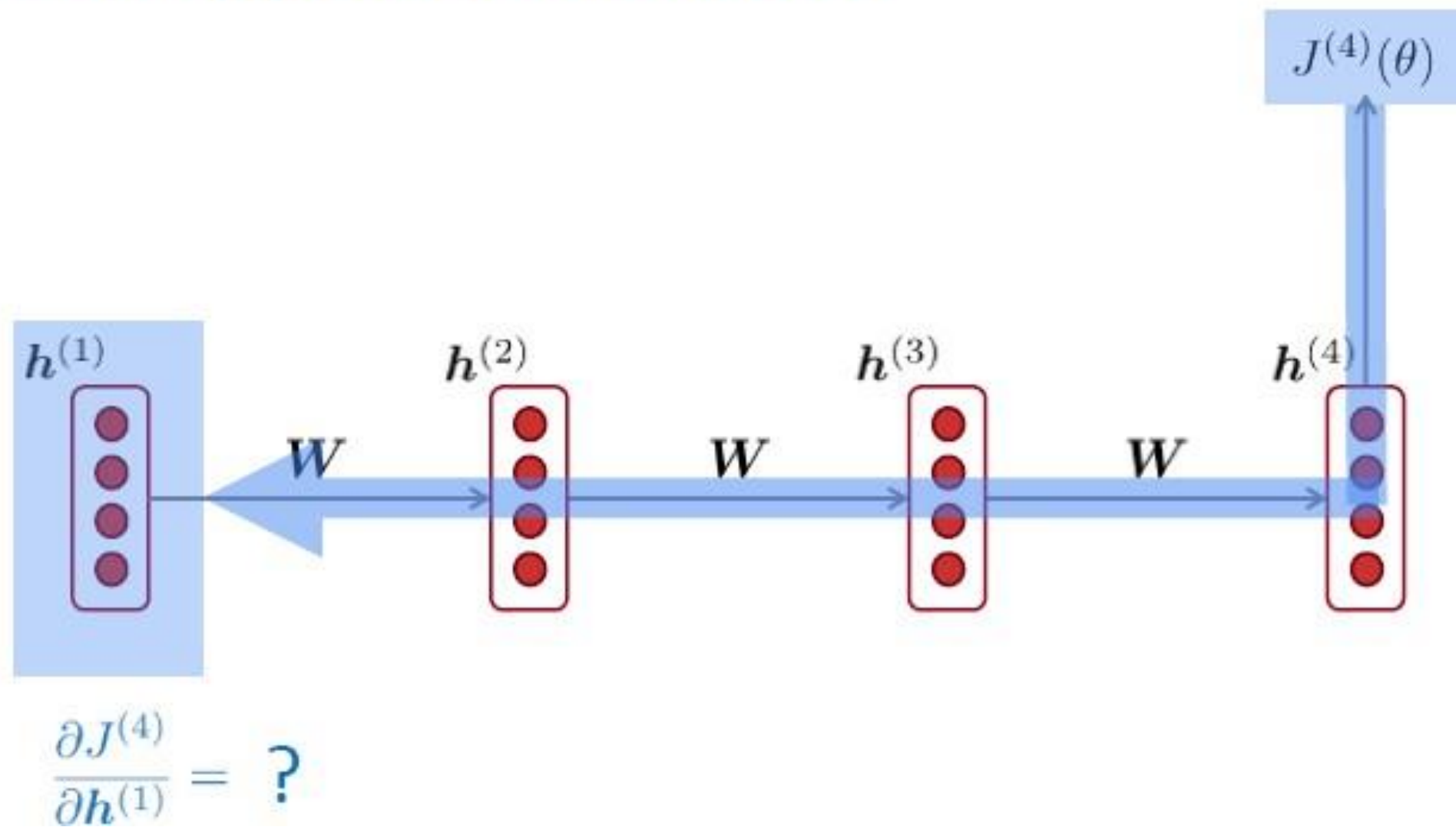
$$\frac{\partial h_{15}}{\partial h_2} = \frac{\partial h_{15}}{\partial h_{14}} \frac{\partial h_{14}}{\partial h_{13}} \cdots \frac{\partial h_{\cancel{2}}^3}{\partial h_{\cancel{4}_2}}$$

That's a lot of chain rule! These chains of gradients are troublesome because if less than 1 they can cause the loss from the word shepherd with respect to the word brown to approach 0, thereby *vanishing*. This makes it difficult for the weights to take into account words that occur at the start of a long sequence. So the word "brown" when doing a forward propagation, may not have any effect in the prediction of "shepherd" because the weights weren't updated due to the vanishing gradient. This is one of the major disadvantages of RNNs.

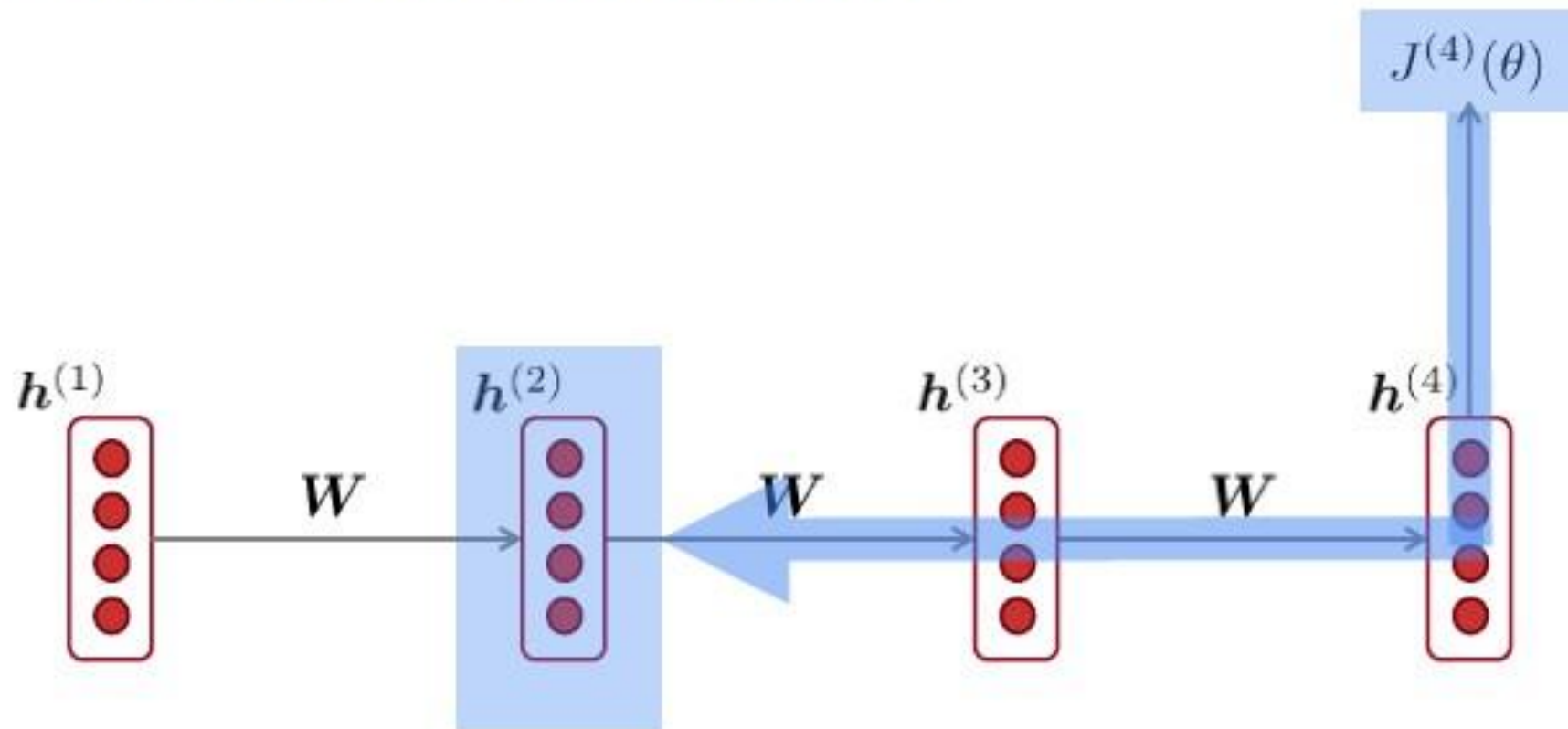
Vanishing gradient intuition



Vanishing gradient intuition



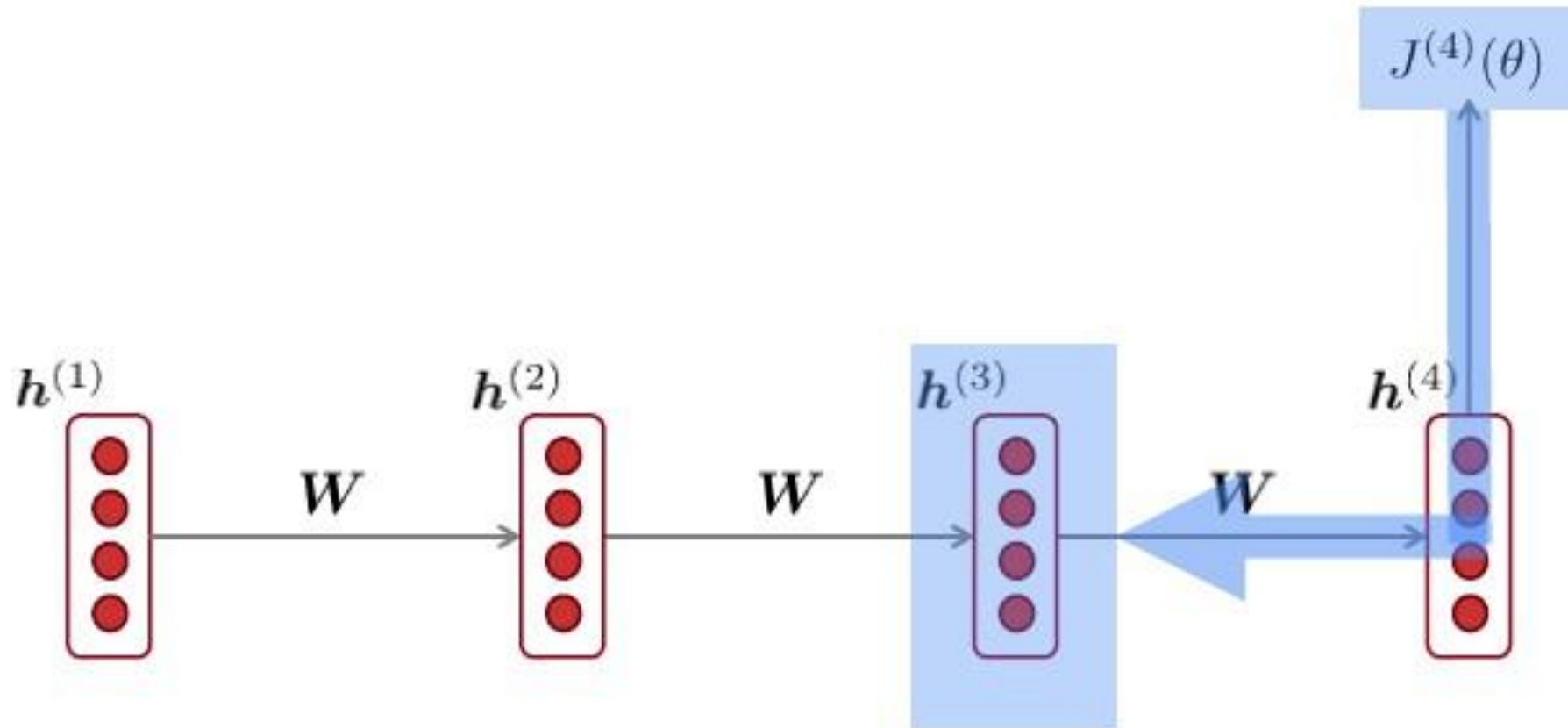
Vanishing gradient intuition



$$\frac{\partial J^{(4)}}{\partial h^{(1)}} = \frac{\partial h^{(2)}}{\partial h^{(1)}} \times \frac{\partial J^{(4)}}{\partial h^{(2)}}$$

chain rule!

Vanishing gradient intuition

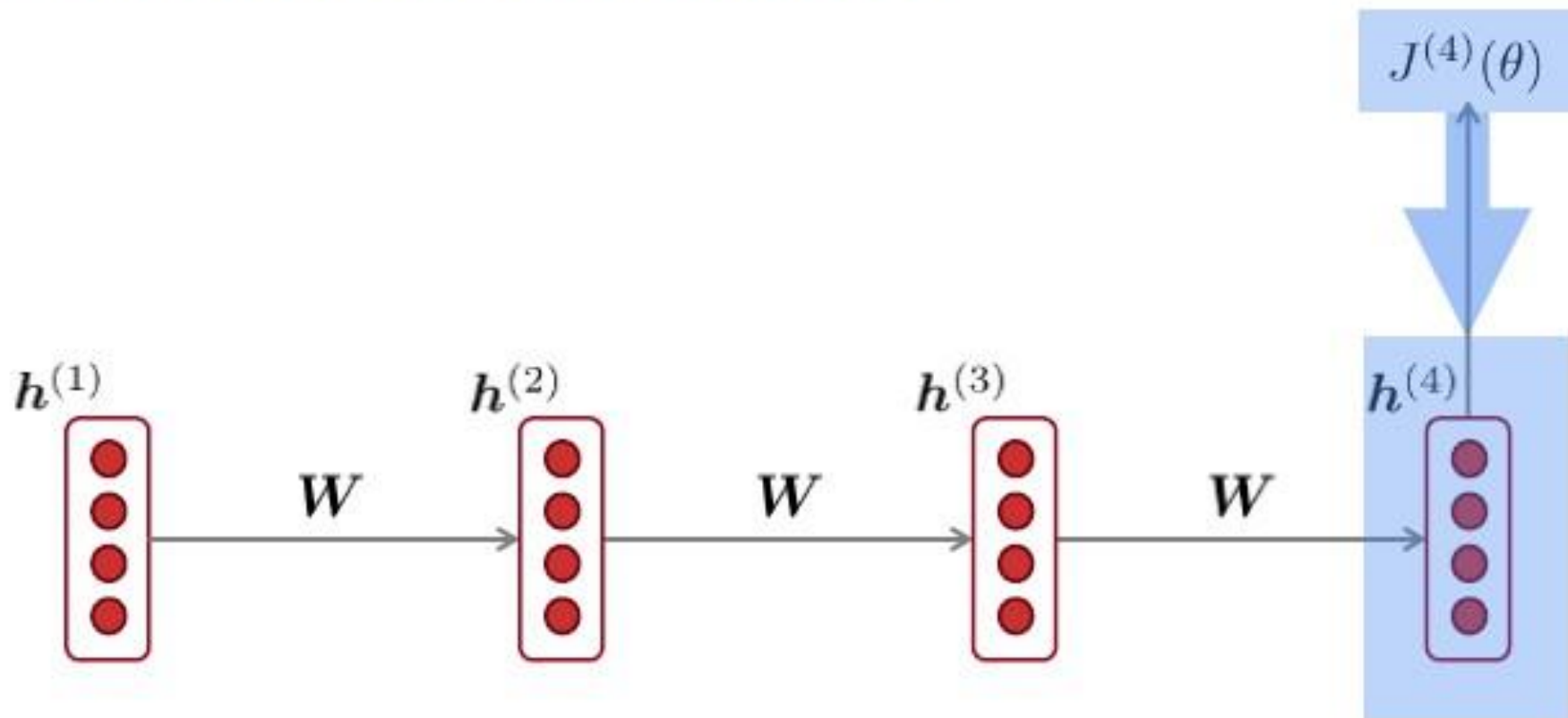


$$\frac{\partial J^{(4)}}{\partial h^{(1)}} = \frac{\partial h^{(2)}}{\partial h^{(1)}} \times \dots$$

$$\frac{\partial h^{(3)}}{\partial h^{(2)}} \times \frac{\partial J^{(4)}}{\partial h^{(3)}}$$

chain rule!

Vanishing gradient intuition



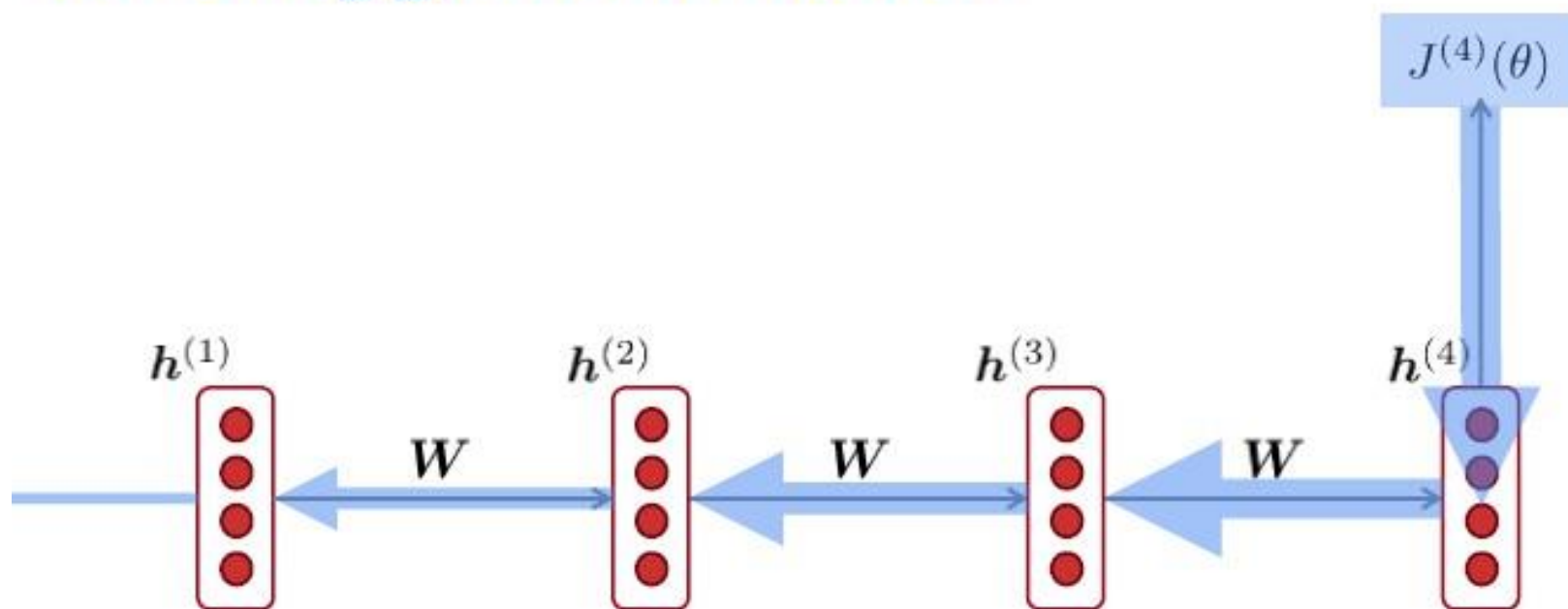
$$\frac{\partial J^{(4)}}{\partial h^{(1)}} = \frac{\partial h^{(2)}}{\partial h^{(1)}} \times$$

$$\frac{\partial h^{(3)}}{\partial h^{(2)}} \times$$

$$\frac{\partial h^{(4)}}{\partial h^{(3)}} \times \frac{\partial J^{(4)}}{\partial h^{(4)}}$$

chain rule!

Vanishing gradient intuition

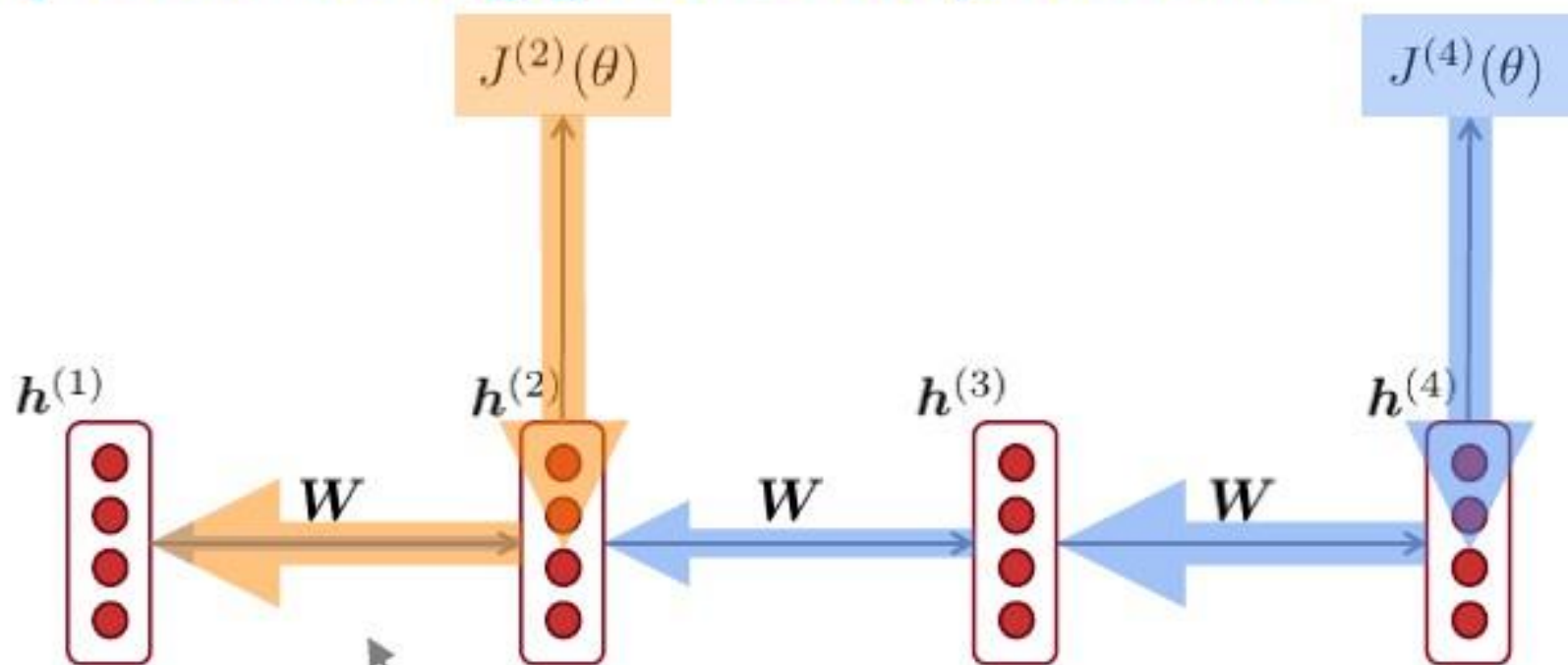


$$\frac{\partial J^{(4)}}{\partial h^{(1)}} = \boxed{\frac{\partial h^{(2)}}{\partial h^{(1)}}} \times \boxed{\frac{\partial h^{(3)}}{\partial h^{(2)}}} \times \boxed{\frac{\partial h^{(4)}}{\partial h^{(3)}}} \times \frac{\partial J^{(4)}}{\partial h^{(4)}}$$

What happens if these are small?

Vanishing gradient problem:
When these are small, the gradient signal gets smaller and smaller as it backpropagates further

Why is vanishing gradient a problem?



Gradient signal from faraway is lost because it's much smaller than gradient signal from close-by.

So model weights are only updated only with respect to near effects, not long-term effects.

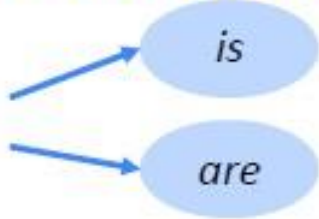
Why is vanishing gradient a problem?

- Another explanation: Gradient can be viewed as a measure of *the effect of the past on the future*
- If the gradient becomes vanishingly small over longer distances (step t to step $t+n$), then we can't tell whether:
 1. There's *no dependency* between step t and $t+n$ in the data
 2. We have *wrong parameters* to capture the true dependency between t and $t+n$

Effect of vanishing gradient on RNN-LM

- **LM task:** *When she tried to print her tickets, she found that the printer was out of toner. She went to the stationery store to buy more toner. It was very overpriced. After installing the toner into the printer, she finally printed her _____*
- To learn from this training example, the RNN-LM needs to **model the dependency** between “tickets” on the 7th step and the target word “tickets” at the end.
- But if gradient is small, the model **can't learn this dependency**
 - So the model is **unable to predict similar long-distance dependencies** at test time

Effect of vanishing gradient on RNN-LM

- **LM task:** *The writer of the books ____* 
- **Correct answer:** *The writer of the books is planning a sequel*
- **Syntactic recency:** *The writer of the books is* (correct) 
- **Sequential recency:** *The writer of the books are* (incorrect) 
- Due to vanishing gradient, RNN-LMs are better at learning from **sequential recency** than **syntactic recency**, so they make this type of error more often than we'd like [Linzen et al 2016]

How to fix vanishing gradient problem?

- The main problem is that *it's too difficult for the RNN to learn to preserve information over many timesteps.*

- In a vanilla RNN, the hidden state is constantly being rewritten

$$\mathbf{h}^{(t)} = \sigma \left(\mathbf{W}_h \mathbf{h}^{(t-1)} + \mathbf{W}_x \mathbf{x}^{(t)} + \mathbf{b} \right)$$

- How about a RNN with separate memory?

Vanishing Gradient Problem solution

- GRU
- LSTM

Why is exploding gradient a problem?

- If the gradient becomes too big, then the SGD update step becomes too big:

$$\theta^{new} = \theta^{old} - \overbrace{\alpha}^{\text{learning rate}} \underbrace{\nabla_{\theta} J(\theta)}_{\text{gradient}}$$

If the weights or derivatives are larger than one, multiplying them repeatedly causes the gradient values to grow exponentially into massive numbers. Huge weight adjustments cause parameters to jump wildly. The loss function fails to converge and instead increases erratically.

- This can cause **bad updates**: we take too large a step and reach a bad parameter configuration (with large loss)
- In the worst case, this will result in **Inf** or **NaN** in your network (then you have to restart training from an earlier checkpoint)

Gradient clipping: solution for exploding gradient

- Gradient clipping: if the norm of the gradient is greater than some threshold, scale it down before applying SGD update

Algorithm 1 Pseudo-code for norm clipping

$\hat{\mathbf{g}} \leftarrow \frac{\partial \mathcal{E}}{\partial \theta}$
if $\|\hat{\mathbf{g}}\| \geq \textit{threshold}$ **then**
 $\hat{\mathbf{g}} \leftarrow \frac{\textit{threshold}}{\|\hat{\mathbf{g}}\|} \hat{\mathbf{g}}$
end if

- Intuition: take a step in the same direction, but a smaller step

Reading

- Chapter 9, Speech and Language Processing. Daniel Jurafsky & James H. Martin. Third edition

<https://web.stanford.edu/~jurafsky/slp3/9.pdf>